

SOFTWARE

Software Classification

Software is simply a collection of instructions executed by a computer's processor. It is broadly split into two distinct categories:

- **System Software:** Programs that manage, maintain, or operate the computer hardware itself (e.g., Operating Systems and Utilities).
- **Application Software:** Programs that allow you to do specific user tasks, like creating a document or playing a game.

Hardware Context

- **Single-Purpose Computers:** Built to run only one piece of software (e.g., a microwave controller).
 - **General-Purpose Computers:** Built to run a vast range of different software applications (e.g., tablets, smartphones, and laptops).
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Utility Software

Utility software is a type of system software that carries out specific configuration, maintenance, and optimization tasks.

1. Backup Utilities

- Creates a duplicate copy of files and applications.
- **Exam Point:** They are usually scheduled to run automatically at off-peak times (like late at night) so they do not slow the system down while a user is working.

2. Defragmentation Utilities

- **The Problem:** Over time, files stored on a mechanical hard disk drive (HDD) become **fragmented**—broken into pieces and scattered out of order across the drive. The read/write head has to search multiple spots, which slows down file loading times.
- **The Solution:** Defragmentation utilities reorder these data blocks so that pieces of the same file are stored right next to each other. This speeds up file access time.

3. Compression Utilities

- Reduces the physical storage size of files.

- **How it works:** It spots repeating data blocks. For example, instead of writing out data as eight **a**s, six **b**s, and two **c**s, an algorithm like **Run-Length Encoding (RLE)** compresses it to a smaller notation: **a8b6c2**.
- **Decompressing:** Compressed files cannot be read directly by their original programs, so utility software must decompress (extract) them first.

4. Disk Formatting Utilities

- Prepares a storage drive (like a USB stick or hard drive) for its very first use by laying down a file system framework.
- **Warning:** Formatting an already used drive marks all existing data as unreadable. While formatting doesn't completely erase data from existence immediately (specialist recovery tools can sometimes find it), it makes the space available to be overwritten.

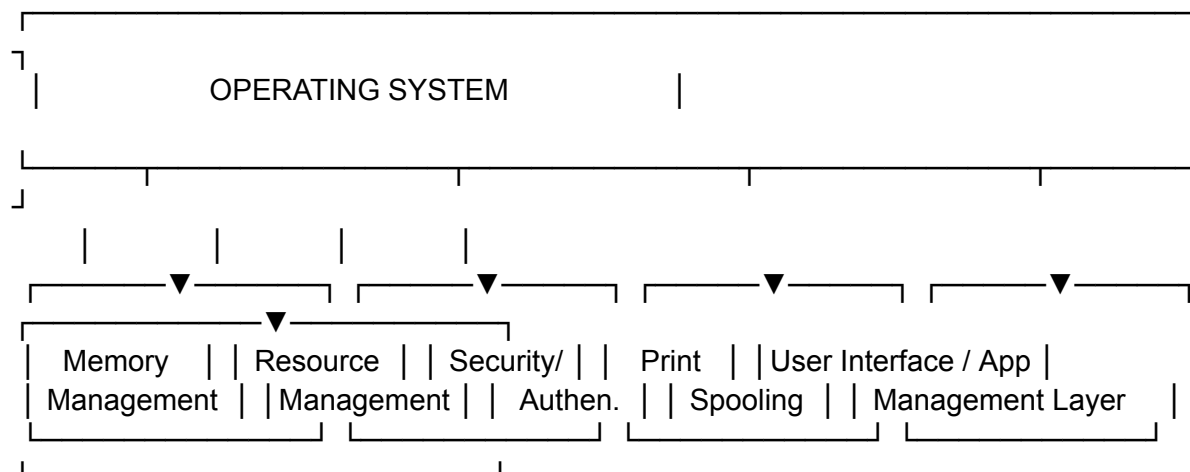
Operating Systems (OS)

The Operating System is the primary software layer that lets you control and manage the computer's underlying hardware.

Core Variations

- **Single-User OS:** Only supports a single general user profile. It cannot be customized for individual accounts and is common in consumer household appliances.
- **Network OS:** Manages multi-user login authentication, separates files so users cannot peek into each other's accounts, and coordinates access to shared network folders and network printers.

The 5 Essential Functions of an OS



1. **Memory Management:** Allocates the precise amount of temporary storage (RAM) an application needs to run. When you close the app, the OS reclaims that space for other programs.
2. **Resource Management:** Allocates processor (CPU) attention and internal hardware (like a graphics card) to various software programs. If two apps demand a printer at once, the OS queues the requests safely.
3. **Security (Authentication):** Controls machine access via usernames, passwords, biometrics (fingerprints), or smart cards. It also manages software firewalls to block unauthorized external network connections.
4. **Print Spooling:** Computers send documents to a printer much faster than the physical printer hardware can ink the pages. The OS puts these pages into an orderly queue (a buffer) so your computer doesn't freeze waiting for the printer to finish.
5. **Application Management & Interface:** Coordinates how application software requests talk to the physical hardware.

Application Software Classification

 **EXAM TIP:** When answering exam questions, always refer to an application by its generic **software type** (e.g., *spreadsheet software*) rather than using specific commercial brand names (e.g., *Microsoft Excel*).

Generic Software Type	Primary Focus & Core Features	Examples Mentioned
Word Processing	Documents that are primarily text-based. Includes spelling/grammar checkers, change tracking, and simple layout formatting.	LibreOffice Writer
Desktop Publishing (DTP)	Built for complex, precise page layouts (e.g., magazines or multi-page books).	Adobe InDesign
Spreadsheets	Automatic calculations using formulas and functions. Used for financial modeling and exploring "what-if" scenarios.	Microsoft Excel
Databases (DBMS)	Structured data storage. Features data entry forms, query editors for specific lookups, and clean report builders.	Oracle MySQL
Web Authoring	Creating web pages that export cleanly into HTML code. Often uses WYSIWYG ("What You See Is What You Get") visual environments.	Adobe Dreamweaver

Graphics Editors	Creating and adjusting Vector graphics (images made mathematically of shapes, points, and lines).	Serif Affinity Designer
Photo Editors	Modifying Bitmap images (digital photos made of grids of pixels). Features include cropping, red-eye removal, and brightness tuning.	GIMP
Sound Editors	Cutting/joining audio, adjusting track volumes, applying noise reduction, or altering the file's tempo (playback speed).	Audacity
Presentation Software	Combining text, images, video, and animation into sequential pathways or slides to support public speaking.	Prezi
Control Software	Reads inputs from physical electronic sensors, processes a logic choice, and sends a command out to physical mechanical actuators/motors.	Lift doors, robotic arms, smart heating
Project Management	Arranging dependent multi-stage tasks into timelines (Gantt charts) to pinpoint milestones and locate the Critical Path .	Gantt charting software
Communication Software	Providing remote network connection entry points to swap data. Includes Web Browsers, Email Clients/Webmail, SMS, MMS, and Instant Messaging.	Web browsers, TweetDeck

Software Licensing & Patches

Software Licences

A software licence is a legal contract giving an individual permission to install and activate an application on a computer processor. Using unlicensed software is a legal violation known as software piracy.

- **Free or Open-Source Software:** Grants users the legal freedom to copy, study, alter, and share the program. The underlying text instructions—the **Source Code**—is completely visible to the public.
 - *Note:* "Free" means free from restrictions on use, not necessarily free of cost. Authors can technically charge a fee to distribute it.
- **Proprietary Software:** Owned and copyrighted by an entity under a specific brand identity. The source code is locked away and kept secret. If a proprietary program is given away entirely for free, it is called **Freeware**.

Why Software Updates Are Vital

Developers push update patches through internet servers for five main reasons:

1. Fixing critical security exploits and code vulnerabilities.
2. Eliminating runtime system errors (bugs).
3. Ensuring compatibility with newly launched Operating Systems.
4. Improving computing performance and code execution efficiency.
5. Adding useful new tools or improving overall usability.

 **System Rule:** Always make a full data backup *before* running software updates in case an unexpected conflict causes software dependencies to break.

CHAPTER QUESTIONS

- 1 State what is meant by the term 'application software'. (1)
- 2 State three types of utility software. (3)
- 3 Describe the role of the operating system. (2)
- 4 Explain why utility software allows the user to schedule backup processes. (2)
- 5 Explain the benefit of using compression utilities to archive files. (2)
- 6 Describe how presentation software can help a presenter to give a talk. (2)
- 7 Describe one use of control software in a relevant industry. (2)
- 8 Discuss why a building company might use project management software. (8)
- 9 State three types of service provided by communication software. (3)
- 10 State two reasons for updating software. (2)